

BIOS detects and executes the boot loader software located in the boot device, which is capable of loading the kernel, thus giving control to the PC.

The boot loader phase: The completion of self-tests initialises the master boot record or MBR, located on the first sector of the bootable disk, which is very small. GRUB is responsible for the execution of the kernel and the *initrd* images in the file system. The MBR executes the GRUB boot loader, which allows us to choose between the multiple kernel images installed on the machine. GRUB then loads the user-selected kernel and passes control on to it. The GRUB entry and default entry time can be changed by editing the *grub.cfg* file or using any third party application.

The kernel phase: Once the control is passed to the kernel, it acts as an intermediate layer or mediator between the software and hardware of a computer. The kernel, being one of the most fundamental components of the operating system, handles all system processes such as I/O request management, task scheduling, memory management and overall system control. The first process that is executed by the kernel is the *init* process.

Operating system: The *init* process's first job is to check the system's properties, the file system and hardware to make sure these are running fine. *Init* executes each run-level one after the other and the corresponding scripts. Based on the run-level, scripts are executed to start or stop various processes to make sure the system is functional. Once the *init* process completes all the tasks, the control is given back to the kernel.

Installing GRUB in Ubuntu

Although GRUB is automatically incorporated into a normal Linux installation, there are times when we need to install it in various storage devices as per the user's convenience during multiple booting. If users want to install Linux in a specific partition, they are supposed to specify the drive on which to embed the boot loader information.

Entire drive: While installing Ubuntu using the 'Erase disk and install Ubuntu' option, the user chooses where Ubuntu and GRUB will be installed, automatically.

Something else: In this method, users are given the extra option to install the boot loader into any device that they choose. This is needed when there is an OS already installed in the system or is going to be installed. The disk where BIOS is set up to boot or the partition where Linux is installed is generally selected in this case.

Making a GRUB bootable CD-ROM: When booting becomes difficult due to some compatibility issues, a GRUB bootable CD-ROM can be created. So there is no need to create a floppy or hard disk image file. Booting from a CD-ROM involves using a special image file, *cdboot.img*, which is included in the *core.img* file. Also, *grub.cfg*, the main configuration file, should be incorporated with the other image files.

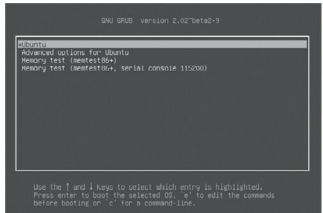


Figure 1: Normal GRUB



Figure 2: Grub Customizer - list configuration

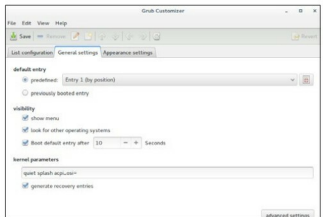


Figure 3: Grub Customizer - general settings

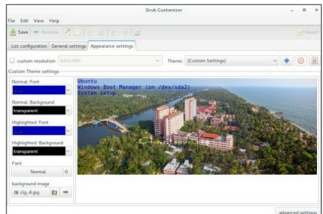


Figure 4: Grub Customizer - appearance settings